1.1 Exercises

September 15, 2023

1

Parabola.

2

Given a quadratic function in the form $y = ax^2 + bx + c$, if $a > 0$ then the y-coordinate of the vertex is the minimum value, and if $a < 0$, it is the maximum value.

3

$y = 4x^2$

x	-2	-1	0	1	2
y	16	4	0	4	16

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The x-coordinate of the vertex is given by $-\frac{b}{2a}$, so the x-coordinate acutally should be $-\frac{24}{2(4)} = -3$.

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The value of c is -7, not 7.

33

Because a is negative, there is a maximum value. The x-coordinate of the vertex is $-\frac{0}{2(-6)} = 0$. So, the y-coordinate of the vertex is $-6(0)^2 - 1 = -1$, which is also the maximum value of this function.

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Because a is positive, there is a minimum value. The x-coordinate of the vertex is $-\frac{8}{2(2)} = -2$. So, the y-coordinate of the vertex is $2(-2)^2 + 8(-2) + 7 = -1$, which is also the minimum value of this function.

37

Because a is positive, there is a minimum value. The x-coordinate of the vertex is $-\frac{6}{2(\frac{3}{2})} = -2$. Then, the y-coordinate of the vertex is $\frac{3}{2}(-2)^2+6(-2)+4 = -2$, which is also the minimum value of this function.

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Because a is negative, there is a maximum value. The x-coordinate of the vertex is $-\frac{-7}{-2(\frac{1}{4})} = -14$. Then, the y-coordinate of the vertex is $-\frac{1}{4}(-14)^2 - 7(-14) + 2 = 51$, which is also the maximum value of this function.

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D

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C

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$a = -0.02,\ b = 1,\ c = 6$

42

$a = -0.01,\ b = 0.7,\ c = 6$

43

$y = x^2 - 8x + 3$, $y = x^2 - 8x + 10$, $y = x^2 - 8x - 12$

44

C

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A

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B

53

In a parabola, points on the parabola with the same y-coordinate are equidistant from the axis of symmetry. We are given two points $(2, 3),\ (-4, 3)$, which both have the same y-coordinates. So, they lie equidistant from the axis of symmetry. The midpoint of the line formed between these two points is $(-1,\ 3)$, and hence, the equation of the axis of symmetry is $x = -1$.

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We are given a function $y = ax^2 + bx + c$. The x-coordinate of the vertex of this function is $-\frac{b}{2a}$. Then, the y-coordinate of the vertex is:
$a(-\frac{b}{2a})^2 + b(-\frac{b}{2a}) + c$
$\rightarrow a(\frac{b^2}{4a^2}) - \frac{b^2}{2a} + c$
$\rightarrow \frac{b^2}{4a} - \frac{b^2}{2a} + c$
$\rightarrow \frac{b^2(2a)}{4a(2a)} - \frac{b^2(4a)}{2a(4a)} + c$
$\rightarrow \frac{b^2(2a) - b^2(4a)}{8a} + c$
$\rightarrow \frac{b^2(-2a)}{8a} + c$
$\rightarrow -\frac{b^2}{4a} + c$

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This can be solved by finding the vertex of the function. First, we simplify the function:
$y = 4000 - 80x + 200x - 4x^2$
$\rightarrow y = -4x^2 + 120x + 4000$
The x-coordinate of the vertex is $-\frac{120}{2(-4)} = 15$. Then, the y-coordinate of the vertex is $-4(15)^2 + 120(15) + 4000 = 4900$. So, the maximum revenue comes if the price is increased by \$0.75, leading to a total sale of $\$4,900$.

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We can represent this situation with the model for $R(x)$ the monthly revenue:
$R(x) = (320 - 20x) \times (70 + 5x)$
$\rightarrow = 22400 + 1600x - 1400x - 100x^2$

$\rightarrow = -100x^2 - 200x + 22400$ Then, the x-coordinate of the vertex is $-\frac{-200}{2(-100)} = -1$. Thus, the y-coordinate of the vertex is 22500. Hence, the shop should reduce the price by \$20 to produce maximum monthly revenue.

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This can be solved by finding the minimum value, that is, the y-coordinate of the vertex. The x-coordinate of the vertex is:
$-\frac{-\frac{7}{15}}{2(\frac{1}{9000})}) = 2100$, and hence, the y-coordinate of the vertex is $\frac{1}{9000}(2100)^2 - \frac{7}{15}(2100) + 500 = 10$. Thus, at it lowest point, the cable is 10 ft. above the road.

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If it could, the maximum value of this jump would be greater than 3. Well, the x-coordinate of the vertex of this function is $-\frac{1.3}{2(-0.2)} = 3.25$. Then, the y-coordinate of the vertex is $-0.2(3.25)^2 + 1.3(3.25) = 2.1125$. Hence, because the maximum value of this graph is less than 3, the moust cannot jump over a 3 ft. high fence.

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(a): For the earth, the function is: $y = -\frac{32}{10000}x^2 + x$ while the equation for the moon is: $y = -\frac{5.3}{10000}x^2 + x$.
(b): After graphing these two functions, it can be seen that the golf ball travels 312.5 ft. on earth, and 1886.792 ft. on the moon. (c): The ratio of the distances traveled is $\frac{1886.792}{312.5} \approx 6.04$. We can see that $\frac{32}{5.3} \approx 6.04$. Hence, it can be inferred that, if $F(g_1, x_1) = y, F(g_2, x_2) = y$ then $\frac{x_1}{x_2} = \frac{g_2}{g_1}$.

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We can set up the equations:
$A = wl,\ P = 2w + l$ This yields:
$l = P - 2w$. Hence, we can multiplly both sides by w, and then substitue A in: $A = w(P - 2w) = -2w^2 + Pw$. Then, we find the vertex of this parabola. The x-coordinate of the vertex is $-\frac{P}{2(-2)} = \frac{P}{4}$. Hence, the y-coordinate of the vertex

is $A = -2(\frac{P}{4})^2 + P(\frac{P}{4})$
$\rightarrow = -2(\frac{P^2}{16}) + \frac{P^2}{4}$
$\rightarrow \frac{P^2}{-8} + \frac{P^2}{4}$